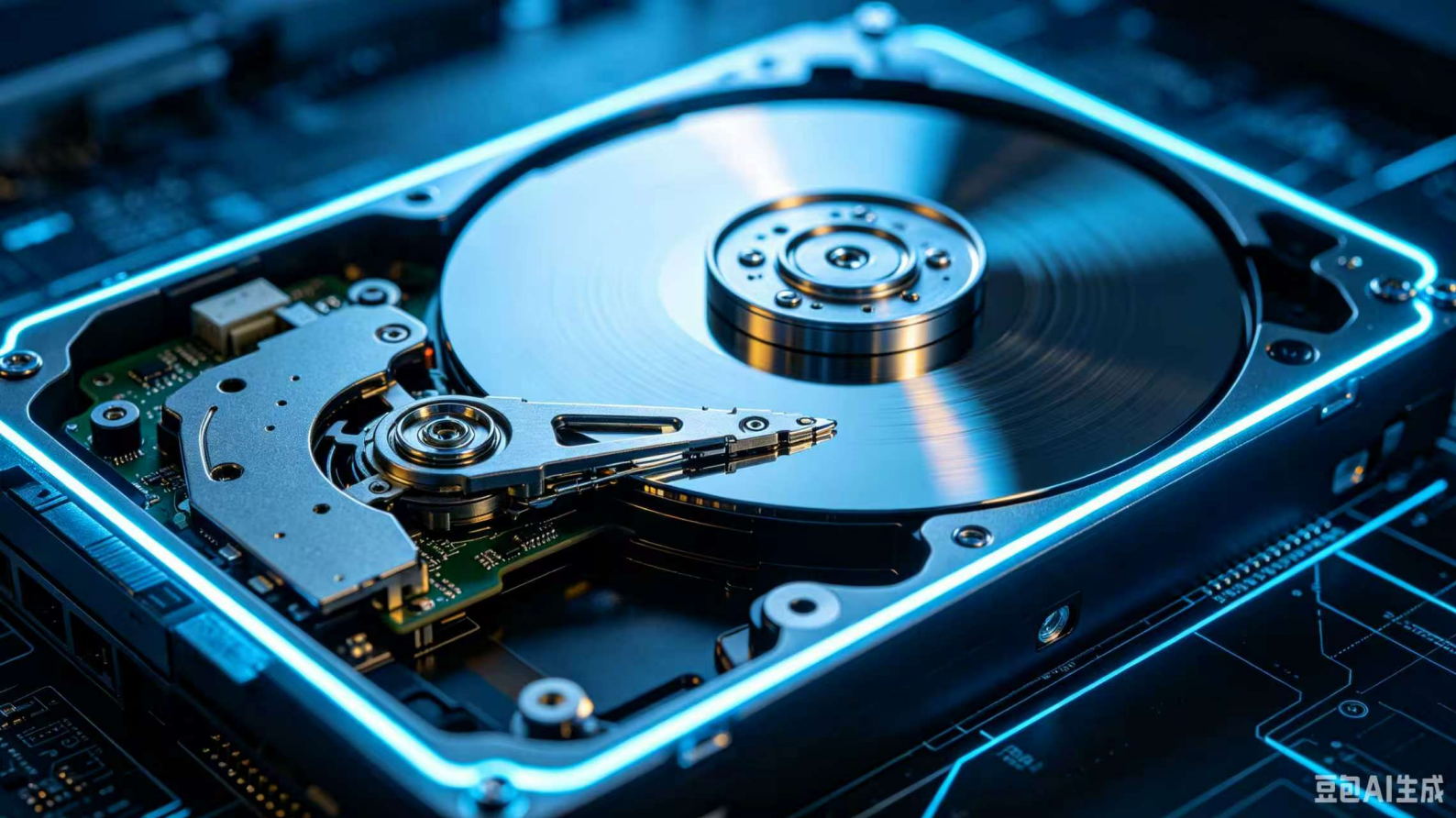




3. Electrostatic Potential

I think most people entering research find that by far the most difficult question is where to start, especially when confronted with something that is actually new.
– P. W. Anderson

In classical mechanics, the law of conservation of mechanical energy can greatly reduce the calculation effort. However, when the system is charged, electromagnetic interactions must be taken into consideration as well. When charged objects move, the electric forces between them in general do work. If a potential energy can be defined for these forces, as it can for gravity, energy conservation will simplify such calculations greatly.

3.1 Potential Energy Difference

The formula of Coulomb's law for point charges is:

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

which resembles the formula of gravitational force between mass points:

$$\mathbf{F}_{12} = -G \frac{m_1 m_2}{r_{12}^2} \hat{r}_{12}$$

In the above equations, $\mathbf{F}_{12}$ means the force from object 1 onto object 2, whereas $\hat{r}_{12}$ is the unit vector from object 1 to object 2. The overall negative sign of gravitational force means it's attractive, whereas the Coulomb force can be either positive or negative, depending on the signs of q_1 and/or q_2 .

Recall that the gravitational force is conservative, so that one can assign a potential energy to celestial systems. The similarities between Coulomb's law and gravitational force suggest potential energy can also be defined for a charge system, and that electrostatic forces are also conservative.

In fact, if we consider a point charge Q located at the origin, and another point charge q moves around Q from $\mathbf{r}_1$ to $\mathbf{r}_2$, the work done by the electrostatic force is

$$W = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{r} \cdot d\mathbf{r}$$

It is easy to verify that $\hat{r} \cdot d\mathbf{r} = dr$ (please try it yourself). Therefore

$$W = \frac{1}{4\pi\epsilon_0} qQ \int_{r_1}^{r_2} \frac{dr}{r^2} = \frac{1}{4\pi\epsilon_0} qQ \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

which depends only on the initial and final positions and is irrelevant to the detailed path it takes.

We now consider a charged object O that creates a static electric field $\mathbf{E}(\mathbf{r})$ in the space. Again, suppose the object is fixed at the origin, and another point charge q moves around it from one position to another. If the point charge q is sufficiently small such that its presence would not induce charge redistribution on O , the electrostatic field $\mathbf{E}(\mathbf{r})$ will not vary in time. In this case, we can always imagine "slicing" O into infinitesimal charged particles, each with fixed position and fixed charge. Thus, the work done by the electrostatic force also depends only on the initial and final positions, meaning that electrostatic force is conservative, or:

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0$$

for electrostatic field.

It is therefore also straightforward to calculate the potential energy through the work done by the electrostatic force.

1. First, we need a reference configuration for zero potential energy. One can choose this arbitrarily, but for finite charge systems, it is conventional to use a configuration in which the charged objects are infinitely separated as the reference.
2. Next, we bring the charged objects together, calculate the work W done by the electrostatic force.
3. Finally, the potential energy is given by $U = -W$.

Suppose a test point charge q_0 moves from initial position i to final position f within an arbitrary electrostatic field $\mathbf{E}(\mathbf{r})$. The work done can be written as:

$$W = \int_i^f \mathbf{F} \cdot d\mathbf{r} = \int_i^f q_0 \mathbf{E} \cdot d\mathbf{r} = q_0 \int_i^f \mathbf{E} \cdot d\mathbf{r}$$

Accordingly, the electric potential energy difference is given by:

$$\Delta U = U_f - U_i = -W = -q_0 \int_i^f \mathbf{E} \cdot d\mathbf{r}$$

Taking the reference point $\mathbf{O}$, chosen freely, as the zero of potential energy, the potential energy at $\mathbf{r}$ is then

$$U(\mathbf{r}) = q_0 \int_{\mathbf{r}}^{\mathbf{O}} \mathbf{E}(\mathbf{r}') \cdot d\mathbf{r}'$$

3.2 Electric Potential

In the above equation, it appears that the electric potential energy U is proportional to the test charge q_0 . In fact, the integral

$$V(\mathbf{r}) = \int_{\mathbf{r}}^{\mathbf{O}} \mathbf{E}(\mathbf{r}') \cdot d\mathbf{r}' \quad (3.1)$$

has nothing to do with q_0 , as it is assumed that the test charge does not change the charge distribution of the source of the electrostatic field $\mathbf{E}$. Thus, $V(\mathbf{r})$ is a property of the electrostatic field $\mathbf{E}$ only, and is called **electric potential**. Please be careful to distinguish it from the **electric potential energy**. Formally, it can be explained as **electric potential energy per unit charge**.

With the above definition, we can calculate the electric potential by 1) calculating the electrostatic field $\mathbf{E}(\mathbf{r})$; 2) defining a zero-potential reference point $\mathbf{O}$; 3) choosing a path and doing the integral. Since electrostatic force is conservative, any path should be equivalent.

Unlike the electric field $\mathbf{E}(\mathbf{r})$, the electric potential $V(\mathbf{r})$ is a scalar function of spatial position $\mathbf{r}$. Writing the potential difference between two points along a path as $V(\mathbf{r}') - V(\mathbf{r}) = - \int_{\mathbf{r}}^{\mathbf{r}'} \mathbf{E} \cdot d\mathbf{l}$, we have $|V(\mathbf{r}') - V(\mathbf{r})| \leq \int |\mathbf{E}| dl$, so V is continuous in space as long as $\mathbf{E}$ stays bounded along that path. This continuity survives even where $\mathbf{E}$ itself jumps, as at the surface of a charged spherical shell or of an infinite charged plane: V is continuous across the surface, and only its gradient jumps. That means $V(\mathbf{r}) = V_0$ defines an “equipotential surface” (or isosurface of electric potential). By definition, no net work is done by electric field when a charged particle moves between two points on the same equipotential surface. It is also apparent that the electric field vector always points from higher potential isosurface to lower potential isosurface. In fact, the electric field lines are always perpendicular to equipotential surfaces. This can be easily derived from its definition.

3.2.1 Discrete Charge Systems

For a single particle of charge q located at the origin, it is fairly simple to derive its electric potential $V(\mathbf{r})$. Assuming the zero-potential reference point at infinity, we have:

$$V(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E}(\mathbf{r}') \cdot d\mathbf{r}'$$

Putting in $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$ and $\hat{\mathbf{r}} \cdot d\mathbf{r} = dr$, we have

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad (3.2)$$

Note that r is the norm of $\mathbf{r}$, and can be viewed as the distance from charge q to the position for which the electric potential V is calculated.

Thus, if there is a group of n charges, with charges $q_1, q_2, \dots, q_n$ at distances $r_1, r_2, \dots, r_n$ away from the position for which V is calculated, the potential will be (taking the zero-potential reference point at infinity as well)

$$V = \sum_{i=1}^n V_i = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$$

Or more explicitly, if these charges are located at $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n$, and we calculate V at position $\mathbf{r}$, then:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{|\mathbf{r} - \mathbf{r}_i|} \quad (3.3)$$

because $r_i = |\mathbf{r} - \mathbf{r}_i|$.

Example 3.1: Electric potential for a dipole.

Let's suppose the dipole consists of a positive charge $+q$ at $\mathbf{d}/2$ and negative charge $-q$ at $-\mathbf{d}/2$, and that we calculate the potential at $\mathbf{r}$, and $d \ll r$.

Method 1

Using the above equation for a discrete charge system, we have:

$$\begin{aligned}
 V(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \left[\frac{+q}{|\mathbf{r} - \mathbf{d}/2|} + \frac{-q}{|\mathbf{r} + \mathbf{d}/2|} \right] \\
 &= \frac{q}{4\pi\epsilon_0} \frac{|\mathbf{r} + \mathbf{d}/2| - |\mathbf{r} - \mathbf{d}/2|}{|\mathbf{r} - \mathbf{d}/2| |\mathbf{r} + \mathbf{d}/2|} \\
 &\approx \frac{q}{4\pi\epsilon_0} \frac{\mathbf{d} \cdot \hat{\mathbf{r}}}{r^2} \\
 &= \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \mathbf{r}}{r^3} \tag{3.4}
 \end{aligned}$$

Method 2

Alternatively, we can start from the electric field of a dipole obtained in Section 1.4 and integrate it directly. Take the path from infinity to $\mathbf{r}$ along the radial direction, so that the polar angle θ between $\mathbf{p}$ and $\mathbf{r}$ is fixed, $d\mathbf{l} = dr \hat{\mathbf{r}}$, and

$$\mathbf{E} \cdot \hat{\mathbf{r}} = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}] \cdot \hat{\mathbf{r}} = \frac{1}{4\pi\epsilon_0} \frac{2(\mathbf{p} \cdot \hat{\mathbf{r}})}{r^3} = \frac{1}{4\pi\epsilon_0} \frac{2p \cos \theta}{r^3}$$

Therefore

$$\begin{aligned}
 V(\mathbf{r}) &= - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l} = - \frac{2p \cos \theta}{4\pi\epsilon_0} \int_{\infty}^r \frac{dr'}{r'^3} \\
 &= - \frac{2p \cos \theta}{4\pi\epsilon_0} \left(-\frac{1}{2r^2} \right) = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \mathbf{r}}{r^3}
 \end{aligned}$$

which agrees with Method 1.

3.2.2 Continuous Charge Systems

For finite size continuous charge systems, we can slice them into small charge elements, each of which can be regarded as a point charge, and sum them together. When we slice them into infinitely many pieces, the sum in the above formula becomes an integral. Therefore, formally we can write:

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$$

or more explicitly,

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{dq(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

where $dq(\mathbf{r}')$ means the charge element at position $\mathbf{r}'$, normally obtained as the charge density times the volume element $dq(\mathbf{r}') = \rho(\mathbf{r}')d^3\mathbf{r}'$. In the former equation, you must distinguish r , which means the distance between dq and the point where V is calculated, and the position vector itself (that's why V does not have spatial parameters in the former equation).

However, the above method has a caveat. If the continuous charge system extends to infinity and does not fall off fast enough, the above integral will not converge (please think about the physical reason). In such cases, we have to stick to the original definition and expression.

Example 3.2: Charged rod

Suppose we have a uniformly charged rod of length L , with total charge Q . Calculate the electric potential at point P , located a distance x from one end O of the rod, with PO perpendicular to the rod.

Without loss of generality, we choose x axis along OP , y axis along the rod. We take a small slice of the rod from y to $y + dy$, the slice has charge:

$$dq = \frac{Q}{L} dy$$

and the distance r between the slice and P is:

$$r = \sqrt{x^2 + y^2}$$

Therefore

$$\begin{aligned} V(x) &= \frac{1}{4\pi\epsilon_0} \int_0^L \frac{\frac{Q}{L} dy}{\sqrt{x^2 + y^2}} \\ &= \frac{Q}{4\pi\epsilon_0 L} \int_0^L \frac{dy}{\sqrt{x^2 + y^2}} \\ &= \frac{Q}{4\pi\epsilon_0 L} \ln \left(\frac{L + \sqrt{L^2 + x^2}}{x} \right) \end{aligned}$$

In the above example, if we assume that L is small, or $L \ll x$, we can also approximate the final result by expanding it. The integration:

$$\begin{aligned} \int_0^L \frac{dy}{\sqrt{x^2 + y^2}} &= \int_0^{L/x} \frac{dt}{(1 + t^2)^{1/2}} = \int_0^{L/x} \left(1 - \frac{1}{2}t^2 + \dots \right) dt \\ &= \frac{L}{x} - \frac{1}{6} \left(\frac{L}{x} \right)^3 + \dots \end{aligned}$$

thus, the potential can be written as (keeping the leading correction)

$$V(x) = \frac{1}{4\pi\epsilon_0} \frac{Q}{x} - \frac{1}{24\pi\epsilon_0} \frac{QL^2}{x^3}$$

Example 3.3: Charged disk

Suppose we have a uniformly charged disk of radius R , with total charge Q . Calculate the electric potential at point P located on its axis and x away from its center.

Since the system has cylindrical symmetry, we can slice the disk into rings. Points on the same ring have the same distance from P . Suppose one such ring extends from s to $s + ds$; the charge on this ring is proportional to its surface area:

$$dq = \sigma dS = \frac{Q}{\pi R^2} 2\pi s ds = \frac{2Q}{R^2} s ds$$

The distance r between P and any point on this ring is:

$$r = \sqrt{s^2 + x^2}$$

Thus

$$\begin{aligned} V(x) &= \frac{1}{4\pi\epsilon_0} \int_0^R \frac{\frac{2Q}{R^2} s ds}{\sqrt{s^2 + x^2}} \\ &= \frac{Q}{4\pi\epsilon_0 R^2} \int_0^{R^2} \frac{d(s^2)}{\sqrt{s^2 + x^2}} \\ &= \frac{Q}{2\pi\epsilon_0 R^2} \left(\sqrt{R^2 + x^2} - x \right) \end{aligned}$$

For $x \gg R$, this reduces to $V(x) \approx \frac{1}{4\pi\epsilon_0} \frac{Q}{x}$, the potential of a point charge, as it must.

In Chapter 1 we obtained the electric field of the same uniformly charged disk (Example 1.1). Once we have the relation between $\mathbf{E}$ and V in Section 3.3, differentiating the potential above with respect to x will reproduce that field.

3.3 From Electric Potential to Electric Field

Except for a free constant corresponding to the choice of zero-potential point, there is indeed a one-to-one correspondence between electric field $\mathbf{E}(\mathbf{r})$ and electric potential $V(\mathbf{r})$. That is to say, given the form of electrostatic field $\mathbf{E}(\mathbf{r})$ in space, there is definite electric potential $V(\mathbf{r})$, and vice versa. While deriving $V(\mathbf{r})$ from $\mathbf{E}(\mathbf{r})$ seems to be straightforward, the reverse is not so obvious: it requires the vector operator ∇ , which we introduce in Chapter 4, where the derivation is given. The result is

$$\mathbf{E} = -\nabla V \quad (3.5)$$

where $\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$ in the Cartesian coordinate system.

3.4 Exercises

- Two point charges are at a distance of $2a$, and each carries a charge q . How much work is needed to bring them to a distance of a ?
- Given the electric potential $V = V_0 \exp[-(x^2 + y^2 + az)/a^2]$, where $V_0 > 0$ is a constant potential and $a > 0$ a constant length, calculate the electric field $\mathbf{E}$ at $\mathbf{r} = (x, y, z)$.
- A thin ring of radius R , centered at the origin and lying in the xy plane, carries a total charge Q distributed uniformly. Take the electric potential to be zero at infinity. Compute the electric field $\mathbf{E}$ and the potential V at the point $(0, 0, z)$ on the z axis, and show that $E_z = -dV/dz$ there. Explain why $\partial V/\partial x$ and $\partial V/\partial y$ vanish on the axis, even though V off the axis depends on x and y .
- Consider a spherical shell with inner radius R_1 and outer radius R_2 , as shown in the figure below. The shell is uniformly charged with a charge density of ρ . Assume that the electric potential is zero at infinity. Calculate the electric potential at any distance r from the origin.

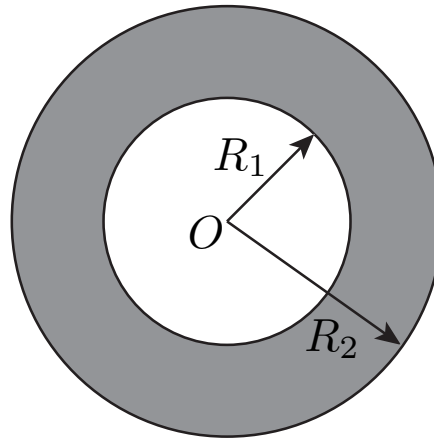


Figure 3.1: A uniformly charged spherical shell.

- Consider two charged point particles, one with charge $-e$ at origin, another with charge ne at $(0, 0, a)$ where $n > 1$. Assume that the electric potential is zero at infinity. Calculate:
 - The electric potential at any given point $\mathbf{r} = (x, y, z)$;
 - Show that the isosurface of zero electric potential is a sphere, and find the center of the sphere.
- Two small particles carry the same charge $q > 0$. Their masses are m and $2m$. They are held at rest a distance d apart, far from any other charge, and then released. Neglect gravity; the only force on each particle is the electrostatic force of the other.
 - Find the speed of each particle when they are very far apart.
 - Suppose instead that the particle of mass $2m$ is held fixed, and only the other particle is released from the same starting position. Find its final speed. Why is it larger than in (a)?

7. Take the electrostatic field to be

$$\mathbf{E}(x, y, z) = \frac{E_0}{a} (y \hat{x} + x \hat{y})$$

everywhere, where the constants $E_0 > 0$ and $a > 0$ are a field strength and a length, and choose $V = 0$ at the origin.

- (a) Find the potential V at the point $P = (x, y, 0)$ by integrating along the path $(0, 0, 0) \rightarrow (x, 0, 0) \rightarrow (x, y, 0)$, and again along the path $(0, 0, 0) \rightarrow (0, y, 0) \rightarrow (x, y, 0)$. Show that the two paths give the same result.
 - (b) Find the equipotential curve in the xy plane that passes through $(a, a, 0)$, keeping only the branch that contains this point. Show that $\mathbf{E}$ is perpendicular to this curve at every point on it.
8. A thin straight rod of length $2L$ carries a uniform linear charge density $\lambda > 0$. The point P lies on the perpendicular bisector of the rod, at a distance s from its midpoint. Take $V = 0$ at infinity.
- (a) Find the potential V at P . Show that, for fixed s , V grows without bound as $L \rightarrow \infty$.
 - (b) Let P_1 and P_2 be two points on the bisector, at distances s_1 and s_2 from the midpoint. Show that $V(P_1) - V(P_2)$ approaches a finite limit as $L \rightarrow \infty$, and find it. Check the result against the field of an infinite line charge, $\lambda/(2\pi\epsilon_0 s)$, which Gauss's law gives.
9. A ball of radius R carries a total charge Q spread uniformly through its volume. Imagine building it by bringing charge slowly in from infinity, one thin spherical layer at a time, so that at every stage the ball is uniformly charged with the final charge density. Take $V = 0$ at infinity.
- (a) When the ball has radius r , find the work needed to bring in the next layer, of thickness dr . Hence show that the total work needed to assemble the ball is $U = \frac{3Q^2}{20\pi\epsilon_0 R}$. Why is U smaller than $QV(R)$, where $V(R)$ is the final potential at the surface?
 - (b) (A numerical estimate.) Suppose the electron were a uniformly charged ball whose rest energy $m_e c^2$ is all of this electrostatic energy. Estimate its radius, using $e = 1.60 \times 10^{-19} \text{ C}$, $m_e c^2 = 8.19 \times 10^{-14} \text{ J}$ and $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$.
10. A point charge $q > 0$ sits at the origin, surrounded by a spherically symmetric distribution of other charge. Together they produce the potential

$$V(r) = \frac{q}{4\pi\epsilon_0} \frac{\exp(-r/a)}{r} \quad (r > 0),$$

where $r = \sqrt{x^2 + y^2 + z^2}$ and $a > 0$ is a constant length; $V = 0$ at infinity.

- (a) Calculate the Cartesian components of $\mathbf{E} = -\nabla V$, and write the result in the form $\mathbf{E} = E(r) \hat{r}$.
- (b) Use Gauss's law to find the total charge $Q(r)$ inside a sphere of radius r centered at the origin. Find $Q(r)$ as $r \rightarrow 0$ and as $r \rightarrow \infty$. What is the total charge of the distribution that surrounds the point charge?